

AGENEWTECH

H1502BQ

Smart Module Specification_V1.3



Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

Wuxi Sunning Technology Co., Ltd. (Abbreviation: Agenew)

No.52, Zone C, Huigu Pioneer Park, Huishan Economic Development Zone, Wuxi

Postal Code: 214174

PhoneNum: +86 0510 8235 0372 Mailbox: Asupport@agenewtech.com

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About the Document

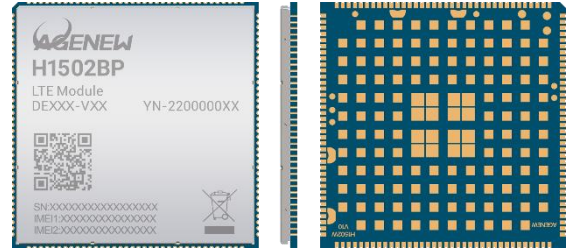
History

Version	Data	Author	Description
-	2024-02-21	Wang Chang	Initial
1.0	2024-02-22	Wang Chang	Controlled Copy
1.1	2024-06-15	Yang Yanlian	Add frequency band allocation
1.2	2024-12-04	Yang Yanlian	Correct a mistake
1.3	2026-03-23	Chen Yun	Change company name

H1502BQ

Multi-network Smart LTE module

Supports Wi-Fi & BT



H1502BQ is a multi-network Smart LTE Cat 4 module based on MediaTek's quad-core ARM Cortex-A53 processor launched by Agenew, equipped with Android operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1502BQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n/ac, and BT5.0 near-field wireless communication. In addition, the H1502BQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The home screen supports HD+ 1280*800@60fps/720*1600@60fps at most. At the same time, H1502BQ supports GPS+GLONASS+BDS+Galileo positioning systems, making its positioning function fast and accurate.

H1502BQ is a patch module with a total of 274 pins, including 146 LCC pins and 128 LGA pins. The size is only 40.5mm × 40.5mm × 2.8mm, which can be embedded in various M2M product applications through pads. H1502BQ integrates rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to HD+ 1280×800@60fps/720×1600@60fps
- ✓ Support 1080P@60fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



quad-core ARM Cortex-A53
64-bit processor



IEEE 802.11
a/b/g/n/ac



Support Android 12



GPS+GLONASS+BDS+Galileo



Powerful multimedia



BT5.0

LTE Cat4		H1502BQ	
Region/Operator	Eurasia/South America		
Operating System	Android 12/Android 13/Android 14		
Memory	2GB+8GB/2GB+32GB/3GB+32GB /4GB+32GB		
Module size (mm)	40.5 × 40.5 × 2.8		
Packaging	LCC + LGA		
Temperature Range			
Operation Temperature	Normal operating temperature: -20 °C ~ + 80°C		
	Storage temperature range: -40 °C ~ + 90 °C		
Frequency Band Information	H1502BQ-E4FB	H1502BQ-E1	H1502BQ-A1
LTE-FDD	B1/B3/B7/B8/B20	B1/B3/B5/B7/B8/B20/B28A/B28B	B1/B2/B3/B4/B5/B7/B8/B28A/B28B
LTE-TDD	B38	B34/B38/B39/B40/B41	B38
WCDMA	B1/B8	B1/B5/B8	B1/B2/B4/B5/B8
GSM/EDGE	B3/B8	B2/B3/B5/B8	B2/B3/B5/B8
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac		
Bluetooth	BT5.0		
GNSS	GPS+GLONASS+BDS+Galileo		
Authentication			
Operator Certification	Undetermined		
Compulsory Certification	Undetermined		
Other Certifications	Undetermined		
Data Rate			
LTE-FDD (Mbps)	150Mbps (DL)/50Mbps (UL)		
LTE-TDD (Mbps)	150Mbps (DL)/50Mbps (UL)		
DC-HSPA+ (Mbps)	42Mbps (DL)/11.2Mbps (UL)		
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)		
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)		
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)		
Outstanding characteristics			
Five-Mode	LTE: Release 12 UMTS: Release 9		
Bandwidth（MHz）	1.4/3/5/10/15/20		
Rx-Diversity	●		
DL supports MIMO 2 × 2	●		
SIM Card Detection	●		
Dual SIM Dual Standby	●		
USB Firmware Upgrade	●		
OTA Firmware Upgrade	●		

Remarks: 1. *: In progress

2. ●: Supports

LTE Cat4	H1502BQ
Electrical specification	
Operating Voltage	3.5~4.4 V, typical 3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	4-lane MIPI_DSI, HD+ 1280×800@60fps
Camera	Support 3 sets of camera, up to 21MP
TP	Capacitive touch screen
Audio	×3 Analog inputs: MIC1, MIC2, HEADMIC (×1 noise cancelling microphone) ×3 Analog outputs: differential linear (requires external audio amplifier), differential receiver and stereo earphone
USB	USB 2.0, Supports OTG, Charge
(U)SIM	2 sets, Supports 1.8/3V (U) SIM card, supports (U) SIM card hot plug detection, dual card dual standby
UART	2 UARTs for debugging and applications
SD Card	1 set
ADC	2 sets, 1 set general ADC interface, 1 set for battery ID
I2C	7 I2Cs to control peripheral devices
SPI	4 SPI interface
PWRKEY	Internal pull up to VBAT
GPIO	18 GPIO reserved
ANT	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna